ALMA-MATER OF QUAID-E-AZAM MOHAMMAD ALI JINNAH

SINDH MADRESSATUL ISLAM UNIVERSITY

Object-Oriented Programming, Spring 2026

Assignment-2

Course Code: **CSC104**
Course Instructor: **Engr. Majid Kaleem**

Submission Deadline: **30/04/26**
Max. Marks: **04**

QUESTION 1:

a) Define *modularity* and *reusability* in the context of software application development.

b) Explain how OOP principles (classes, objects, inheritance, polymorphism, encapsulation) contribute to modularity and reusability.

QUESTION 2:

a) Provide at least two real-world examples where modularity and reusability improve software quality.

b) Describe the concepts of modularity and reusability with the help of the following:

   i. At least two classes with clear responsibilities.

   ii. Use of inheritance and interfaces.

QUESTION 3:

a) Explain the use of control classes, boundary classes, and entity classes in the context of OOP and software design.

b) Describe how each type of class contributes to modularity and reusability.

QUESTION 4:

a) Express by UML class diagrams the interaction between control classes, boundary classes, and entity classes in a small library management system. Label each class type clearly and show how they collaborate to support modularity and reusability.

G☹☺d Luck!